

Radiologic-Radiomic Machine Learning Models for Differentiation of Benign and Malignant Solid Renal Masses: Comparison With Expert-Level Radiologists

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OBJECTIVE. The objective of our study was to compare the performance of radiologic-radiomic machine learning (ML) models and expert-level radiologists for differentiation of benign and malignant solid renal masses using contrast-enhanced CT examinations.

MATERIALS AND METHODS. This retrospective study included a cohort of 254 renal cell carcinomas (RCCs) (190 clear cell RCCs [ccRCCs], 38 chromophobe RCCs [chrRCCs], and 26 papillary RCCs [pRCCs]), 26 fat-poor angioleiomyolipomas, and 10 oncocytomas with preoperative CT examinations. Lesions identified by four expert-level radiologists (> 3000 genitourinary CT and MRI studies) were manually segmented for radiologic-radiomic analysis. Disease-specific support vector machine radiologic-radiomic ML models for classification of renal masses were trained and validated using a 10-fold cross-validation. Performance values for the expert-level radiologists and radiologic-radiomic ML models were compared using the McNemar test.

RESULTS. The performance values for the four radiologists were as follows: sensitivity of 73.7–96.8% (median, 84.5%; variance, 122.7%) and specificity of 48.4–71.9% (median, 61.8%; variance, 161.6%) for differentiating ccRCCs from pRCCs and chrRCCs; sensitivity of 73.7–96.8% (median, 84.5%; variance, 122.7%) and specificity of 52.8–88.9% for differentiating ccRCCs from fat-poor angioleiomyolipomas and oncocytomas (median, 80.6%; variance, 269.1%); and sensitivity of 28.1–60.9% (median, 84.5%; variance, 122.7%) and specificity of 75.0–88.9% for differentiating pRCCs and chrRCCs from fat-poor angioleiomyolipomas and oncocytomas (median, 50.0%; variance, 191.1%). After a 10-fold cross-validation, the radiologic-radiomic ML model yielded the following performance values for differentiating ccRCCs from pRCCs and chrRCCs, ccRCCs from fat-poor angioleiomyolipomas and oncocytomas, and pRCCs and chrRCCs from fat-poor angioleiomyolipomas and oncocytomas: a sensitivity of 90.0%, 86.3%, and 73.4% and a specificity of 89.1%, 83.3%, and 91.7%, respectively.

CONCLUSION. Expert-level radiologists had obviously large variances in performance for differentiating benign from malignant solid renal masses. Radiologic-radiomic ML can be a potential way to improve interreader concordance and performance.

Renal cell carcinoma (RCC) constitutes approximately 70% of all renal cancers [1, 2]. According to the 2016 World Health Organization classification [3], the most common subtypes of RCCs are clear cell RCCs (ccRCCs), papillary RCCs (pRCCs), and chromophobe RCCs (chrRCCs), accounting for approximately 70%, 15%, and 5% of all RCCs, respectively. The classification of RCCs into subtypes has become important because each is associated with a different prognosis [3–5]. Oncocytoma, a benign lesion that accounts for 3–5% of renal neoplasms in adults, can be mistaken for RCC at imaging and accounts for 4–10% of nephrectomies in which

RCC is suspected [6]. In addition, up to 5% of renal angioleiomyolipomas have insufficient fat content to be accurately detected using conventional CT and MRI, making them difficult to discriminate from RCC [7]. This ultimately leads to unnecessary surgical treatment of these benign lesions, making them one of the most commonly resected benign renal masses.

CT is gradually evolving into a useful imaging tool in renal mass differential diagnosis [8, 9]. Several studies have reported that the use of enhancement threshold levels could help to distinguish RCC subtypes and discriminate RCC from benign oncocytoma with 77–84% accuracy [6, 10, 11].

Radiologic-Radiomic ML Analysis of Solid Renal Masses

However, the differential diagnosis of renal mass-forming lesions is still difficult, and a variety of imaging findings have been described with different performance results reported. Previously described methods, including region-based or histogram-based measurements of lesion attenuation on multiphase CT or MR images, have shown inconsistent reproducibility [6, 10–14]. Studies have shown that renal masses including different RCC subtypes, angiomyolipomas, and oncocytomas have been described with different gray-level imaging patterns that might be tracked by quantitative texture or radiomic analysis [15, 16]. Radiomics uses machine learning (ML) approaches to extract large-scale imaging parameters, and its potential to improve diagnostic, prognostic, and predictive accuracy is increasingly being studied [13, 16–19]. However, the use of

radiomics as clinical biomarker still necessitates independent tests set in larger cohorts and direct comparison with established radiologic parameters, and expert-level assessment is required to better understand the predictive capabilities of radiomics.

We hypothesized that the characterization of CT-detected renal masses can be improved by radiomics. Therefore, the purpose of this study was to compare the performance of radiologic-radiomic ML models and expert-level radiologists for the differentiation of benign and malignant solid renal masses using contrast-enhanced CT (CECT) examinations.

Materials and Methods

Patients

This retrospective analysis was performed in a cohort of patients undergoing nephrectomy. Ethics committee approval was granted by the local insti-

tutional ethics review board with a waiver of written informed consent. All procedures performed in studies involving human participants were performed in accordance with the 1964 Helsinki Declaration and its later amendments.

We queried our institution's medical records to derive all surgically confirmed cases of renal masses between January 2012 and December 2016. A total of 630 confirmed renal masses were screened in this query: 481 ccRCCs, 41 chrRCCs, 27 pRCCs, 11 oncocytomas, and 70 angiomyolipomas. CT studies of angiomyolipomas were reviewed by two genitourinary radiologists to assess for the presence of macroscopic intral-lesional fat. The absence of macroscopic fat on unenhanced CT images was defined as fat-poor angiomyolipoma. There were 26 pathologically proven renal angiomyolipomas with no macroscopic fat at visual inspection on preoperative CT images. Patients without CECT images, pa-

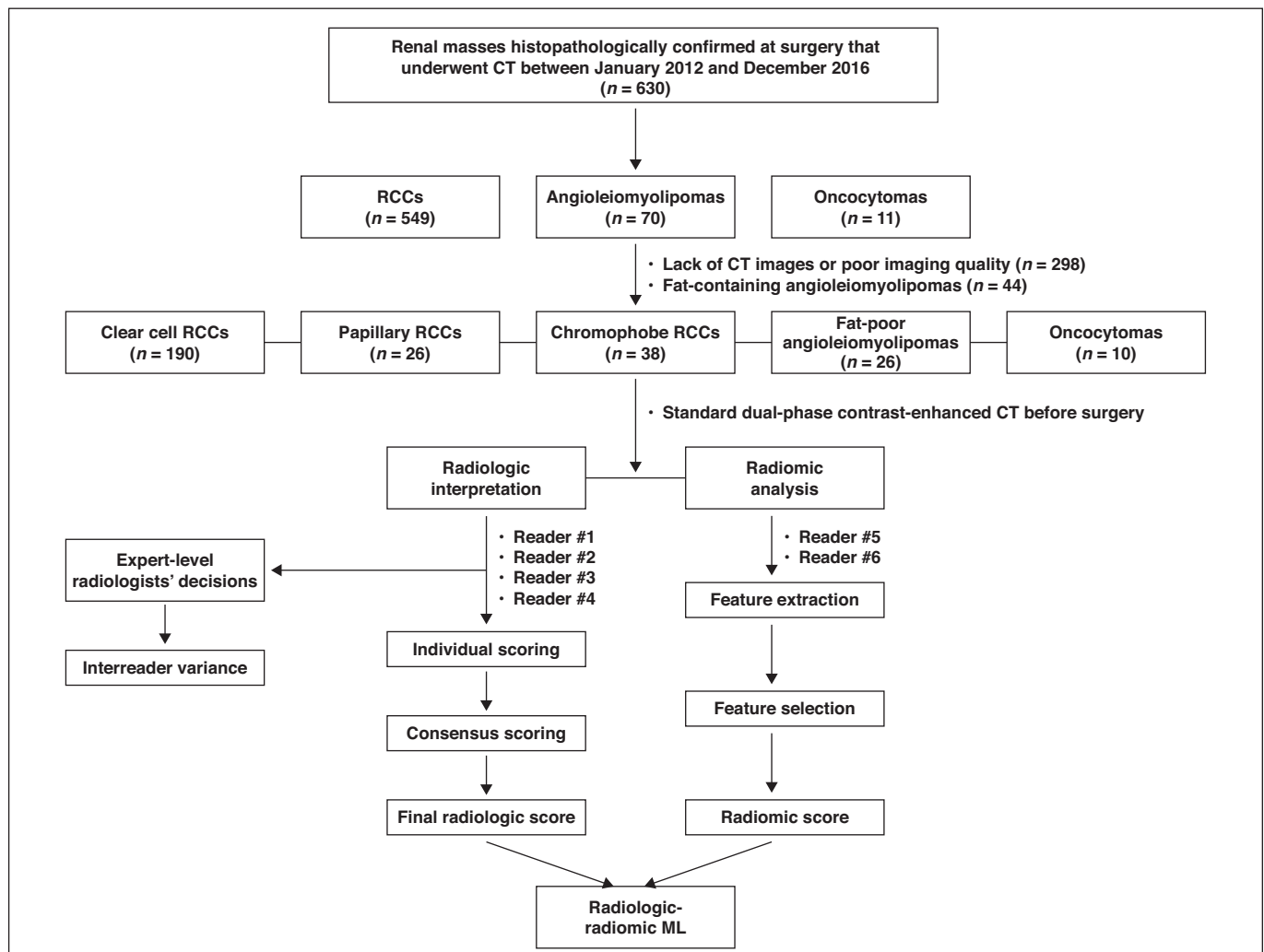


Fig. 1—Study flowchart. Values in parentheses are numbers of lesions. Two patients had two clear cell renal cell carcinomas (RCCs). ML = machine learning.

tients with CT examinations performed at an outside institution, patients with CECT examinations of poor imaging quality, and patients who did not have fat-poor angioleiomyolipomas were excluded. Finally, 252 patients with 254 RCC lesions (190 ccRCCs, 38 chrRCCs, and 26 pRCCs), 10 patients with 10 oncocytomas, and 26 patients with 26 fat-poor angioleiomyolipomas were included in this study (Fig. 1).

CT Examination

All CT examinations were performed on a 128-MDCT scanner (Somatom Definition AS⁺, Siemens Healthcare) at our institution before patients underwent surgical treatment. The CT parameters were as follows: 120 kVp; effective tube current–exposure time product, 180 mAs_{eff}; rotation time, 0.5 second; detector collimation, 32 × 1.2 mm; FOV, 350 × 350 mm; matrix, 512 × 512; and reconstruction section thickness, 1.5 mm. After unenhanced CT, patients underwent standard renal dual-phase (corticomedullary and nephrographic phases) CECT. Nonionic contrast medium (iopromide, Ultravist 300 mg I/mL, Bayer Schering Pharma) was injected via an antecubital vein at a rate of 3.0 mL/s using a CT-compatible power injector for a total volume of 90–120 mL (1.5 mL/kg of body weight). The scanning delay for corticomedullary phase imaging was determined using the scanner's built-in automatic bolus-triggering software (CARE Bolus, Siemens Healthcare). Corticomedullary phase scanning automatically began 10 seconds after the trigger

attenuation threshold (100 HU) was reached at the level of the supraceliac abdominal aorta. At a delay of 40 seconds after corticomedullary phase scanning, nephrographic phase scanning began.

The mean time from CT examination to nephrectomy was 9 days (range, 5–14 days). Of the 288 patients included in this study, 178 were men, and 110 were women. The median age at the time of nephrectomy for all patients was 55 years (age range, 19–85 years).

Radiologic Evaluation

Interpretation of CECT images was performed by four board-certified radiologists (readers 1–4) during clinical routine. All readers are members of the institution's urologic disease management team and had read more than 3000 genitourinary CT and MRI examinations at the time of the study. Before the imaging interpretation, they attended a half-day training session that included studies in the literature that explained in detail the imaging approaches for evaluation of solid renal masses [8].

Imaging interpretations were based on the clinical experience of each radiologist. Generally, readers made decisions about imaging interpretations on the basis of their overall impression and evidence-based knowledge of the CECT features of renal masses. Imaging interpretations included an assessment of likelihood of ccRCC, pRCC, chrRCC, benign fat-poor angioleiomyolipoma, and oncocytoma for each detected lesion. Individual interpretation was designated for the calculation of interobserver variance. Additionally, all

examinations were reviewed using a 2-point scoring scheme by four radiologists after completing a departmental training period. The following radiologic scores were indicated according to methods reported previously [8, 17, 18]: lesion origination (left kidney or right kidney); lesion location (upper or middle or lower pole); maximal cross-sectional diameter on the axial images (Dmax); shape (regular or irregular); margin (smooth or nonsmooth); pseudocapsule (absent or present), which was defined as the fibrous structure formed by the distended growth of the renal mass compressing the surrounding renal parenchyma and appearing as a low-attenuation ring around the tumor on CT images; extrarenal extension (absent; ≤ 50% extension of renal contour; > 50% extension of renal contour); calcification (absent or present); internal hemorrhage (absent or present); internal necrosis (absent or present); heterogeneity (homogeneous or inhomogeneous enhancement at corticomedullary phase and nephrographic phase imaging); internal septa (absent or present); internal arteries (absent or present); and relative enhancement (RE; defined as ratio of CECT intensity between lesion and cortical kidney at corticomedullary phase imaging). Representative cases are provided to introduce typical radiologic features of lesions on CECT images (Fig. 2). All images were scored individually first and then reviewed together after individual evaluation for concordance assurance. Any disagreements between the readers were discussed until a final consensus score was generated.

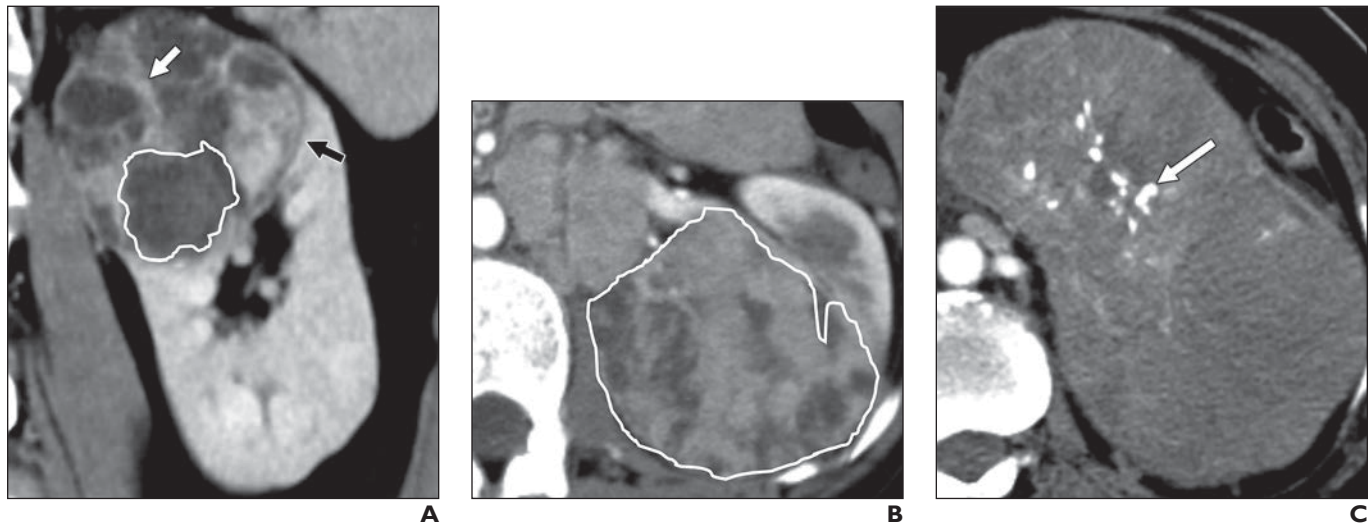


Fig. 2—Representative cases to introduce definitions of radiologic scores on CT images.

A, 68-year-old man with clear cell renal cell carcinoma (ccRCC). Coronal parenchymal image shows ccRCC with well-defined margin and regular shape. In this case, tumor has features that are typical of ccRCC: internal necrosis (outline), internal septa (white arrow), and pseudocapsule (halo sign, black arrow).

B, 36-year-old woman with papillary renal cell carcinoma (pRCC). Axial image shows pRCC (outline) has irregular shape, nonsmooth margin, and inhomogeneous enhancement. Image also shows enlarged lymph nodes in left renal hilum.

C, 47-year-old woman with chromophobe renal cell carcinoma (chrRCC). Axial CT image of chrRCC shows multiple calcifications within tumor (arrow).

(Fig. 2 continues on next page)

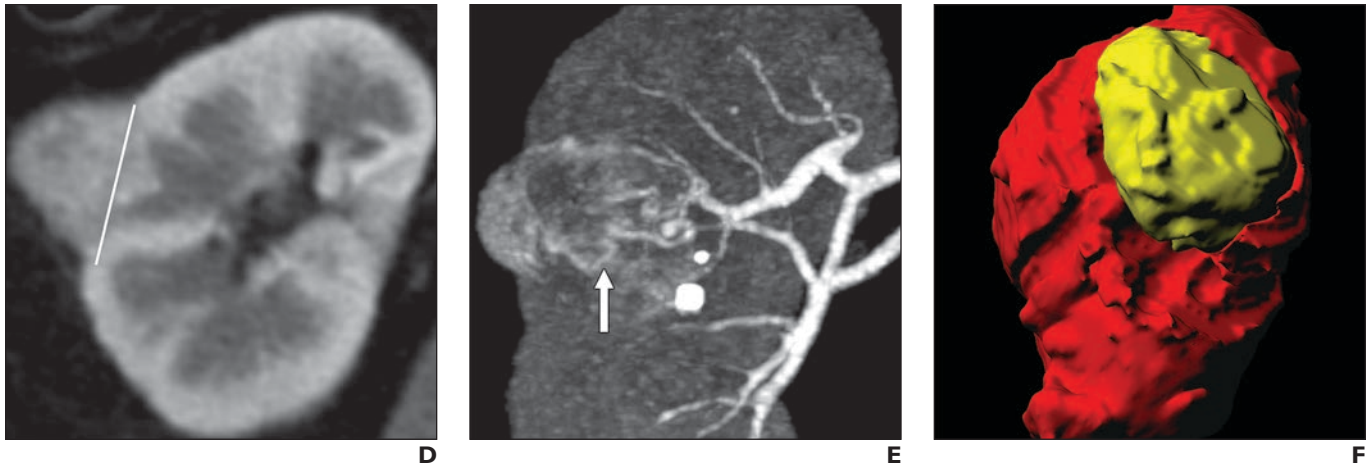


Fig. 2 (continued)—Representative cases to introduce definitions of radiologic scores on CT images.

D, 43-year-old man with clear cell renal cell carcinoma (ccRCC). CT image shows that ccRCC tumor contour results in more than 50% extension of renal contour and that distance to renal sinus is larger than 0.7 cm. Line shows interface between tumor and kidney.

E, 57-year-old man with ccRCC. CT image shows internal arteries sign within ccRCC (arrow).

F, 46-year-old man with ccRCC. Volume-rendering CT image shows semiautomatic volume of interest that was used for radiomic analysis. Red = renal parenchyma, yellow = tumor.

Radiomic Analysis

In the next step, a radiologist resident (reader 5) and an attending radiologist (reader 6) not involved in image interpretation for calculation of expert-level decision performance were asked for radiomic analysis. Lesion segmentation was semi-automatically performed with dedicated in-house software written on Python (version 3.6.1, Python Software Foundation). First, a user-defined ROI was drawn by the radiologist to initially overlay the lesion and perilesional normal tissue; then, a dichotomic classification algorithm was used to automatically segment the lesion from perilesional normal tissue. Finally, the radiologist created a hand-crafted edition of the ROI by outlining precisely the contour of the lesion. The desired contour of the lesion was saved as a 3D image, and the entire volume of interest (VOI) was used for subsequent radiomic analysis.

Previous studies have observed that the accuracy of RCC subtype differentiation with single-phase corticomedullary CECT was comparable to or was even better than that of multiphasic imaging [19, 20]. Therefore, in this study, radiomic features were extracted from 1.5-mm corticomedullary phase CT images using a pyradiomics package on Python software. All processes of radiomic analysis were integrated into an open-source Feature Analysis Explore (FAE) software (Song Y, Zhang J, Yang G) on the GitHub website [21]. Image normalization was performed using a method that remaps the histogram to fit within $\mu \pm 3\sigma$, where μ is gray-level mean of the VOI, and σ is the gray-level SD. Radiomic features of CECT images were computed for each VOI using the five texture analysis methods available in the software package: 19 features of

first-order intensity; 16 features of shape; 16 features of gray-level size zone matrix (GLSZM); 16 features of gray-level run-length matrices (GLRLM), which represent runs of pixels having the same gray-level value; and 26 features of gray-level cooccurrence matrix (GLCM). The extracted radiomic features were normalized to a standard unit with 0-center using the following equation:

$$\bar{x}_n \text{ normalized} = \frac{\bar{x}_n - \bar{x}_n}{\sqrt{x_{1n}^2 + x_{2n}^2 + \dots + x_{mn}^2}},$$

where

$\bar{x}_n$ is the value of feature N ,

and

$\bar{x}_n$ is the mean value of all features.

To evaluate intraobserver reliability, reader 5 repeated the feature extraction twice in a 1-week period, and the readout sessions were conducted over a period of 2 months. The reliability was calculated using the intraclass correlation coefficient (ICC). Radiomic features with both intraobserver and interobserver ICC values greater than 0.9 (indicating excellent stability) were selected for subsequent investigation.

A regression radiomic score was developed to decode the likelihood of ccRCCs, pRCCs and chrRCCs, and fat-poor angioleiomyolipomas and oncocytomas, respectively, with a recursive feature elimination support vector machine (RFE-SVM) method reported previously [22, 23]. In our RFE-SVM, feature selection was integrated with the classifier training, hyperparameter selection, and cross-validation.

The support vector machine (SVM) classifier used radial basis function (RBF) kernels and consisted of the RBF kernel's width parameter (γ), which was defined as $\gamma = \{0.01, 0.02, 0.03, 0.04, \dots, 0.3\}$, and the misclassification penalty (C), which was defined as $C = \{0.1, 0.5, 1, 3, 5, 7, 10\}$. At the end of iteration, the output of the SVM classifier was converted to an adjusted radiomic score (RS):

$$RS = \exp(Y) / 1 + \exp(Y),$$

where Y was the output of RFE-SVM analysis, and the value of radiomic score indicates predictive probability. Because of the relatively low sample size data, especially for data of non-ccRCCs, the radiomic score was trained and validated with a 10-fold cross-validation method that maintained the best compromise between computational cost and reliable estimates.

Radiologic-Radiomic Machine Learning Model and Statistical Analysis

Uni- and multivariate analyses using an adjusted odds ratio (OR) regression model was performed to determine risk factors of clinical endpoints and construct the radiologic-radiomic ML model. Candidate variables were age, sex, 14 consensus radiologic scores, and a 2-point (low or high) radiomic score. The threshold of the 2-point radiomic score was determined using an ROC curve analysis that maximized the Youden index. The statistical analysis was conducted with an R software package (version 3.3.4, The R Foundation for Statistical Computing). The reported statistical significance levels were all two-sided, with statistical significance set at 0.05.

TABLE 1: Characteristics of Patients and Renal Lesions

Characteristic	All Lesions (n = 290)	Clear Cell RCC (n = 190)	Papillary RCC (n = 26)	Chromophobe RCC (n = 38)	Fat-Poor Angioleiomyolipoma (n = 26)	Oncocytoma (n = 10)
Sex						
Male	180 ^a	127 ^a	18	23	7	5
Female	110 ^a	63 ^a	8	15	19	5
Age (y)						
Median (range)	55 (19–85)	59 (23–85)	54 (19–76)	51 (24–83)	47 (24–68)	42 (28–56)
Side						
Right	144	94	14	18	13	5
Left	146	96	12	20	13	5
Surgery						
Partial nephrectomy	196	127	19	20	24	6
Radical nephrectomy	93	63	6	18	2	4
Biopsy	1	0	1	0	0	0
Mean size (cm) ^a	3.94	4.00	4.16	4.61	2.99	3.95
Pathologic stage ^{b,c}						
T1a	192	126	16	22	21	7
T1b	65	41	8	9	4	3
T2	18	13	0	5	0	0
T3	14	10	1	2	1	0
T4	0	0	0	0	0	0

Note—Data are numbers of lesions except where otherwise indicated. RCC = renal cell carcinoma.

^aTwo patients had two clear cell RCCs.

^bDefinitions of pathologic stages are based on the American Joint Committee on Cancer TNM staging system [30].

^cPathologic stage of one papillary RCC lesion was unknown after biopsy.

Results

Of all 290 renal lesions included in this study, 190 (65.5%) were ccRCCs, 26 (9.0%) were pRCCs, 38 (13.1%) were chrRCCs, 26 (9.0%) were fat-poor angioleiomyolipomas, and 10 (3.4%) were oncocytomas at histologic findings. Baseline characteristics for each of the groups are presented in Table 1.

The results of consensus radiologic scores are summarized in Table 2. Statistical differences were observed in six radiologic features among five subgroups with ANOVA test. Regarding the expert-level radiologists' reports, the Cronbach alpha value among the four readers was a mean of 0.64 with a 95% lower confidence limit of 0.58 ($p < 0.05$). The kappa value for comparison of pair of two readers ranged from 0.24 to 0.42 ($p < 0.05$). Overall diagnosis accuracy was 62.1–76.6%.

The results of radiomic analysis are summarized in Table 3. RFE-SVM analysis selected a total of 35 most important features that produced the best accuracy over the validations for the classification of ccRCCs versus pRCCs and chrRCCs, a total of 22 most important features for the classification of ccRCCs versus fat-poor angioleiomyolipomas and oncocytomas, and a total of 11 most important features for the classification of pRCCs and chrRCCs versus fat-poor angioleiomyolipomas and oncocytomas. With the features calculated, the radiomic score reflected significant differences in the three disease groups (Fig. 3). Patients were then classified by radiomic score as having a low radiomic score or high radiomic score in the three groups using a threshold derived from ROC curve analysis.

Eight variables (capsule, heterogeneity, necrosis, internal arteries, septa, extrarenal extension, RE, and radiomic score) were significantly associated with the classification of ccRCCs versus pRCCs and chrRCCs by univariate analysis. At multivariate analysis, the higher RE (> 0.70 ; OR, 35.4; 95% CI, 13.4–93.2) and higher radiomic score (> 0.42 ; OR, 5.5; 95% CI, 2.1–14.0) were independent predictors that were used to form a classification model:

$$Y = -1.56 + 3.57 \times RE + 1.70 \times RS,$$

where RE is relative enhancement and RS is radiomic score.

The resulting radiologic-radiomic ML model showed good accuracy with an area

under the ROC curve (AUC) of 0.93 (95% CI, 0.89–0.95) (Fig. 4A).

Nine variables (age, capsule, heterogeneity, necrosis, internal arteries, Dmax, extrarenal extension, RE, and radiomic score) were significantly associated with the classification of ccRCCs versus fat-poor angioleiomyolipomas and oncocytomas by univariate analysis. At the multivariate analysis, older age (> 40 years; OR, 5.9; 95% CI, 1.7–20.6), presence of capsule (OR, 24; 95% CI, 5.5–104.6), and higher RE (> 0.84 ; OR, 12.9; 95% CI, 4.2–39.8) were independent predictors that were used to form a classification model:

$$Y = -2.96 + 1.78 \times \text{age} + 3.18 \times \text{capsule} + 2.56 \times RE,$$

where RE is relative enhancement.

The resulting radiologic-radiomic ML model showed good accuracy with an AUC of 0.94 (95% CI, 0.89–0.96) (Fig. 4B).

Six variables (sex, capsule, internal arteries, Dmax, RE, and radiomic score) were significantly associated with the classification of pRCCs and chrRCCs versus fat-poor angioleiomyolipomas and oncocytomas by univariate analysis.

Radiologic-Radiomic ML Analysis of Solid Renal Masses

TABLE 2: Radiologic Scores of 290 Renal Lesions Included in Study

Finding	Clear Cell RCC (n = 190)	Papillary RCC (n = 26)	Chromophobe RCC (n = 38)	Fat-Poor Angioleiomyolipoma (n = 26)	Oncocytoma (n = 10)	p
Dmax (cm), mean (SD)	4.6 (2.5)	4.3 (2.4)	4.7 (2.8)	4.1 (1.5)	3.1 (2.2)	0.068
Relative enhancement (AU), mean (SD)	1.02 (0.25)	0.38 (0.13)	0.58 (0.18)	0.79 (0.15)	0.69 (0.17)	< 0.001
Origination						0.992
0, Right kidney	94 (49.5)	14 (53.8)	18 (47.4)	13 (50.0)	5 (50.0)	
1, Left kidney	96 (50.5)	12 (46.2)	20 (52.6)	13 (50.0)	5 (50.0)	
Lesion location						0.290
0, Upper pole	65 (34.2)	6 (23.1)	13 (34.2)	4 (15.4)	5 (50.0)	
1, Middle pole	49 (25.8)	7 (26.9)	14 (36.8)	10 (38.5)	3 (30.0)	
2, Lower pole	76 (40.0)	13 (50.0)	11 (28.9)	12 (46.2)	2 (20.0)	
Lesion shape						0.833
0, Regular	28 (14.7)	6 (23.1)	7 (18.4)	4 (15.4)	2 (20.0)	
1, Irregular	162 (85.3)	20 (76.9)	31 (81.6)	22 (84.6)	8 (80.0)	
Lesion margin						0.842
0, Smooth	167 (87.9)	23 (88.5)	34 (89.5)	23 (88.5)	10 (100)	
1, Nonsmooth	23 (12.1)	3 (11.5)	4 (10.5)	3 (11.5)	0 (0)	
Lesion capsule						< 0.001
0, Absent	63 (33.2)	12 (46.2)	23 (60.5)	24 (92.3)	8 (80.0)	
1, Present	127 (66.8)	14 (53.8)	15 (39.5)	2 (7.7)	2 (20.0)	
Extrarenal extension						0.030
0, Absent	66 (34.7)	12 (46.2)	19 (34.7)	15 (57.7)	4 (40.0)	
1, ≤ 50% Extension	108 (56.8)	12 (46.2)	12 (31.6)	7 (26.9)	6 (60.0)	
2, > 50% Extension	16 (8.4)	2 (7.7)	7 (18.4)	4 (15.4)	0 (0)	
Internal calcification						0.425
0, Absent	171 (90.0)	25 (96.2)	34 (89.5)	26 (100)	9 (90.0)	
1, Present	19 (10.0)	1 (3.8)	4 (10.5)	0 (0)	1 (10.0)	
Internal hemorrhage						0.687
Absent	185 (97.4)	25 (96.2)	38 (100)	26 (100)	10 (100)	
Present	5 (2.6)	1 (3.8)	0 (0)	0 (0)	0 (0)	
Internal necrosis						< 0.001
0, Absent	99 (52.1)	16 (61.5)	31 (81.6)	24 (92.3)	5 (50.0)	
1, Present	91 (47.9)	10 (38.5)	7 (18.4)	2 (7.7)	5 (50.0)	
Lesion heterogeneity						< 0.001
0, Homogeneous	9 (4.7)	12 (46.2)	10 (26.3)	8 (30.8)	0 (0)	
1, Inhomogeneous	181 (95.3)	14 (53.8)	28 (73.7)	18 (69.2)	10 (100)	
Internal septa						0.107
0, Absent	161 (84.7)	22 (87.6)	38 (100)	24 (92.3)	9 (90.0)	
1, Present	29 (15.3)	4 (15.4)	0 (0)	2 (7.7)	1 (10.0)	
Internal arteries						< 0.001
0, Absent	152 (80.0)	26 (100)	17 (44.7)	21 (80.8)	10 (100)	
1, Present	38 (20.0)	0 (0)	21 (55.3)	5 (19.2)	0 (0)	

Note—Data are numbers of lesions with percentages in parentheses, except where otherwise indicated. Continuous data were analyzed by ANOVA test, and counted data were analyzed by Mann-Whitney *U* tests. RCC = renal cell carcinoma, Dmax = maximal cross-sectional diameter on axial images.

TABLE 3: Radiomics Results of Three Groups With Recursive Feature Elimination Support Vector Machine Analysis

Group	No. of Features Selected	Top Three Features			Radiomic Score			Maximum AUC	Radiomic Score Threshold
		1	2	3	Negative Subgroup ^a	Positive Subgroup ^a	<i>p</i>		
Group A ^b	35	First-order uniformity	GLCM informal measure of correlation 2	GLCM maximum probability	0.32 (0.13)	0.56 (0.17)	< 0.001	0.88 (0.14)	> 0.42
Group B ^c	22	GLRLM gray-level nonuniformity normalized	Shape flatness	First-order uniformity	0.56 (0.08)	0.55 (0.07)	0.025	0.73 (0.17)	≤ 0.54
Group C ^d	11	First-order 90th percentile	First-order median	GLCM informal measure of correlation 2	0.79 (0.15)	0.59 (0.17)	< 0.001	0.85 (0.21)	≤ 0.71

Note—Numbers in parentheses are SD. GLCM = gray-level cooccurrence matrix, GLRLM = gray-level run-length matrices.

^aPositive subgroup for groups A and B is composed of clear cell renal cell carcinomas (ccRCCs), and positive subgroup for group C is composed of papillary renal cell carcinomas (pRCCs) and chromophobe renal cell carcinomas (chrRCCs). Negative subgroup is composed of lesions not included in positive subgroup.

^bGroup A is ccRCCs versus pRCCs and chrRCCs.

^cGroup B is ccRCCs versus fat-poor angioleiomyolipomas and oncocytomas.

^dGroup C is pRCCs and chrRCCs versus fat-poor angioleiomyolipomas and oncocytomas.

ate analysis. At the multivariate analysis, presence of capsule (OR, 5.7; 95% CI, 1.5–21.3), lower RE (≤ 0.55; OR, 7.9; 95% CI, 1.9–32.7), and lower radiomic score (≤ 0.71; OR, 5.6; 95% CI, 1.7–17.7) were independent predictors that were used to form a classification model:

$$Y = -1.29 + 1.74 \times \text{capsule} + 2.06 \times \text{RE} + 1.72 \times \text{RS},$$

where RE is relative enhancement and RS is radiomic score.

The resulting radiologic-radiomic ML model showed good accuracy with an AUC of 0.87 (95% CI, 0.79–0.93) (Fig. 4C).

The performance values of four expert-level radiologists were as follows: sensitivity of 73.7–96.8% (median, 84.5%; variance, 122.7%) and specificity of 48.4–71.9% (median, 61.8%; variance, 161.6%) for differentiating ccRCCs from pRCCs and chrRCCs; sensitivity of 73.7–96.8% (median, 84.5%; variance, 122.7%) and specificity of 52.8–88.9% for differentiating ccRCCs from fat-poor angioleiomyolipomas and oncocytomas (median, 80.6%; variance, 269.1%); and sensitivity of 28.1–60.9% (median, 84.5%; variance, 122.7%) and specificity of 75.0–88.9% for differentiating pRCCs and chrRCCs from fat-poor angioleiomyolipomas and oncocytomas (median, 50.0%; variance, 191.1%).

In comparison, the quantitative radiologic-radiomic ML models yielded the following performance values for differentiating ccRCCs from pRCCs and chrRCCs, ccRCCs from fat-poor angioleiomyolipomas and oncocytomas, and pRCCs and chrRCCs from fat-poor angioleiomyolipomas and oncocytomas: a sensitivity of 90.0%, 86.3%, and 73.4% and a specificity of 89.1%, 83.3%, and 91.7%, respectively. The details for comparisons of diagnostic performance of expert-level radiologists, radiologic-radiomic ML, and radiomic score are summarized in Table 4.

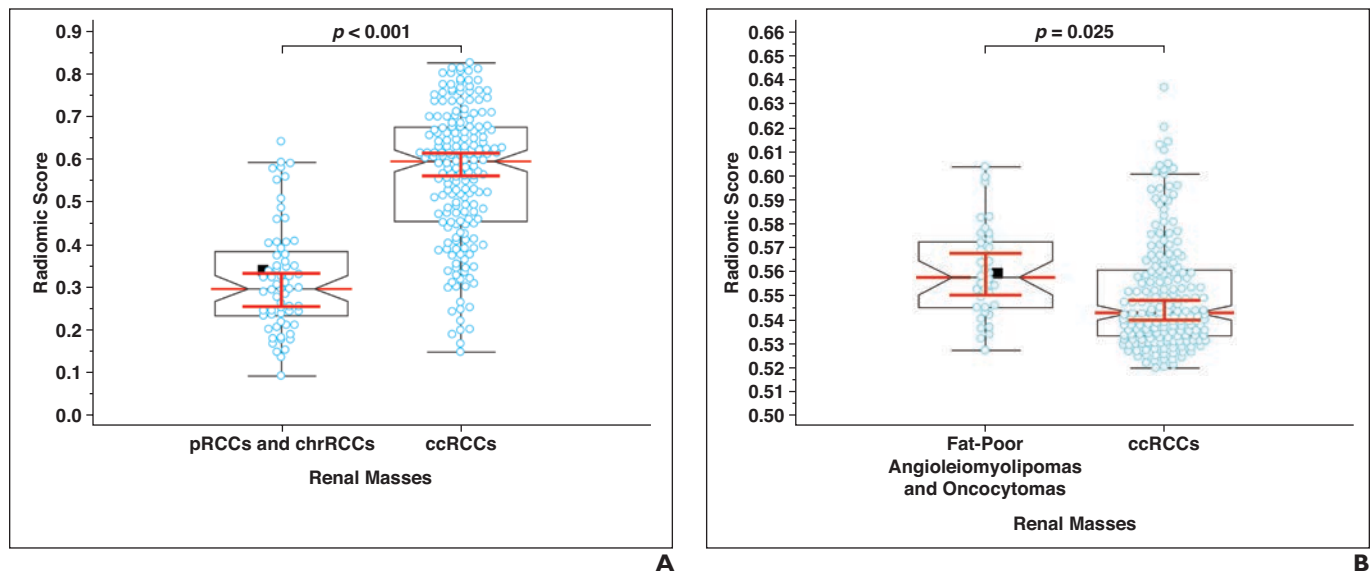
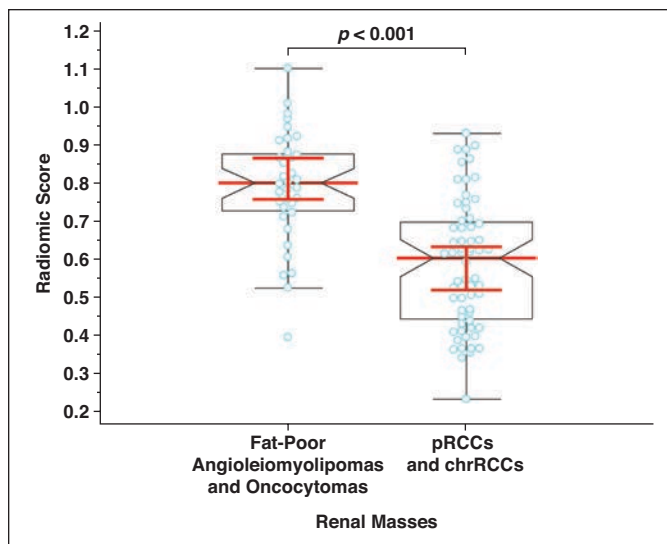


Fig. 3—Notched box-and-whisker distribution plots of radiomic scores in renal masses by subtype.

A–C, Plots show radiomic scores by renal mass subtype. Red lines show median value of radiomic score, red whiskers show corresponding 95% CIs, black notched boxes show 25th and 75th quartiles, and black whiskers show upper and lower limits. pRCCs = papillary renal cell carcinomas, chrRCCs = chromophobe renal cell carcinomas, ccRCCs = clear cell renal cell carcinomas.

(Fig. 3 continues on next page)

Radiologic-Radiomic ML Analysis of Solid Renal Masses



C

Fig. 3 (continued)—Notched box-and-whisker distribution plots of radiomic scores in renal masses by subtype.

A–C, Plots show radiomic scores by renal mass subtype. Red lines show median value of radiomic score, red whiskers show corresponding 95% CIs, black notched boxes show 25th and 75th quartiles, and black whiskers show upper and lower limits. pRCCs = papillary renal cell carcinomas, chrRCCs = chromophobe renal cell carcinomas, ccRCCs = clear cell renal cell carcinomas.

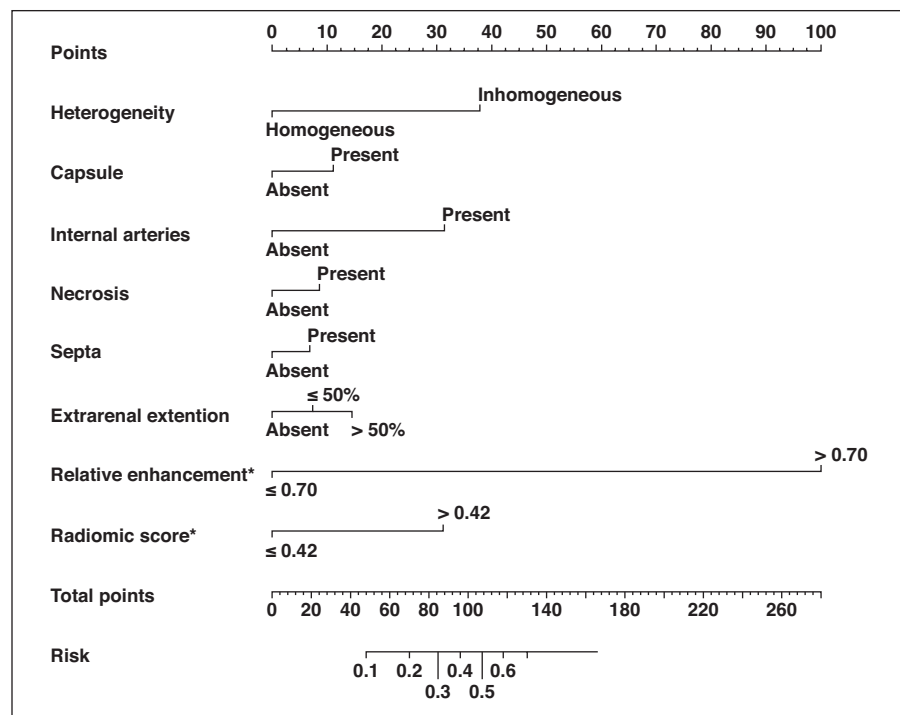
Discussion

Preoperative classification of renal mass subtypes such as ccRCC, pRCC, chrRCC, and benign fat-poor angioleiomyolipoma and oncocytoma is of clinical relevance because renal masses with different biologic characteristics are associated with different treatment choices and postoperative prognosis. Recently, investigators have focused on the application of artificial intelligence such as ML or deep learning (DL) with multiphase CT images in differential diagnosis of renal masses [24–27]. Feng et al. [27] proposed an SVM classifier comprising 42 low-throughput texture features predefined by radiologists' expert knowledge to different small renal tumors with an accuracy of 93.9%. Zhou et al. [24] established DL-extracted high-throughput features and obtained higher accuracy of 97% in a model trained on the ROI dataset when the convolutional neural network weights before mixed6 layer were fixed. The types of tumors in their group led to a higher accuracy because renal cysts and fatty angioleiomyolipomas are significantly different from RCCs on CT images [24]. Coy et al. [25] reported that an TensorFlow Inception (Google) model shows promise as an ML classifier to triage ccRCCs from oncocytomas, with an accuracy of 75.4%, on excretory phase CT with processing mode AEM3. Han et al. [26] trained a deep neural network to classify the subtypes of RCCs (57 clear cell cases, 56 papillary cases, and 56 chromophobe cases), and it showed about 0.85 accuracy. These DL approaches might provide insights regarding unsupervised learning or automatic classification or might yield high

diagnostic performance but are challenged by black-box data (data that are uninterpretable for radiologist) or by loss of information from entire volumetric tumor images as well as from adjacent-slice images. Addi-

tionally, the a priori radiologic identifications such as size, shape margin, and enhancement were not integrated into these DL algorithms and are thus questionable. Therefore, in the present study, we trained our multiobjective models using an interpretable radiologic-radiomic ML algorithm to classify renal mass subtypes using an RFE-SVM analysis. By allowing feature interactions, our RFE-SVM analysis returns a ranking of all features by recursively training the SVM classifier and removing the feature with the smallest ranking score at each iteration. It does not need a large number of training samples to avoid overfitting. Studies have shown that RFE-SVM analysis outperformed or achieved about the same accuracy and performance as the other methods (SVM and adaptive boosting, *t* test) even when not using any augmentation of sample [22, 23, 28].

Preoperative classification of RCC subtypes is of clinical relevance because ccRCC typically has a worse prognosis and a greater likelihood of metastasis than do pRCC and chrRCC; therefore, a noninvasive means of



A

Fig. 4—Performance of radiologic-radiomic machine learning (ML) model for differentiating benign solid renal tumors from renal cell carcinomas (RCCs).

A–C, Graphs show performance of radiologic-radiomic ML model for differentiating clear cell RCCs (ccRCCs) from papillary RCCs (pRCCs) and chromophobe RCCs (chrRCCs) (A); RCCs from fat-poor angioleiomyolipomas and oncocytomas (B); and pRCCs and chrRCCs from fat-poor angioleiomyolipomas and oncocytomas (C). Asterisks indicate independent predictors of multivariate regression analysis. Dmax = maximal cross-sectional diameter on axial images.

(Fig. 4 continues on next page)

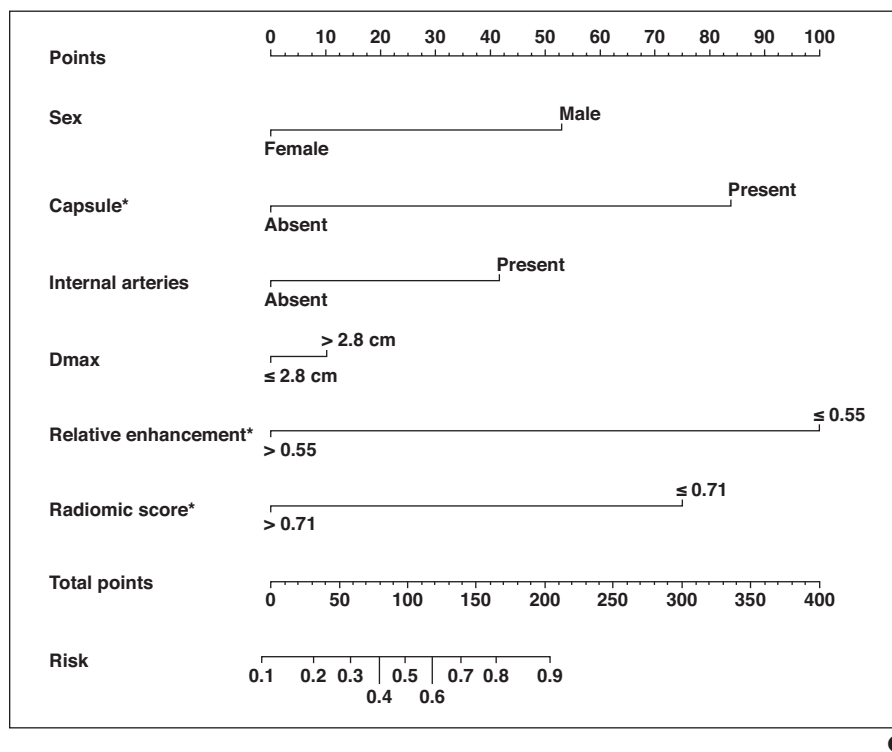
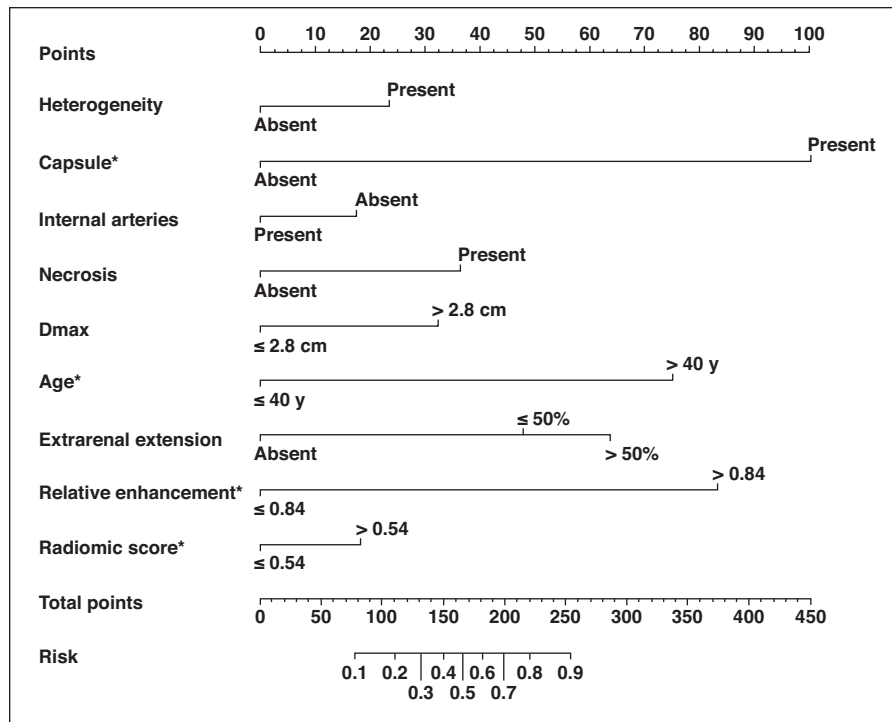


Fig. 4 (continued)—Performance of radiologic-radiomic machine learning (ML) model for differentiating benign solid renal tumors from renal cell carcinomas (RCCs).

A–C, Graphs show performance of radiologic-radiomic ML model for differentiating clear cell RCCs (ccRCCs) from papillary RCCs (pRCCs) and chromophobe RCCs (chrRCCs) (**A**); RCCs from fat-poor angioliomyolipomas and oncocytomas (**B**); and pRCCs and chrRCCs from fat-poor angioliomyolipomas and oncocytomas (**C**). Asterisks indicate independent predictors of multivariate regression analysis. Dmax = maximal cross-sectional diameter on axial images.

discriminating them would be of great importance. Young et al. [6] reported that enhancement at CECT can be used to differentiate ccRCC from pRCC and chrRCC with an accuracy of 85% and 84%, respectively. In our radiologic-radiomic ML model, RE and radiomic score were two independent predictors for disease classification. The integration of RE and radiomic score into the ML model achieved an accuracy of 89.8%, which is higher than the accuracies reported by Young et al. and the accuracy results for our expert-level radiologists (68.1–86.6%). We found that radiomic features, such as first-order uniformity, GLCM informal measure of correlation 2, and GLCM maximum probability, were the top-ranked features of ccRCCs versus pRCCs and chrRCCs. Uniformity is a measure of the homogeneity of the image array, where a greater uniformity implies a greater homogeneity or a smaller range of discrete intensity values. GLCM informal measure of correlation 2 quantifies the complexity of the texture. GLCM maximum probability represents occurrences of the most predominant pair of neighboring intensity values. Compared with pRCCs and chrRCCs, ccRCCs had lower uniformity, higher informal measure of correlation 2 values, and lower maximum probability, showing its inhomogeneous biologic features.

Hodgdon et al. [7] used a similar SVM classifier to analyze texture features at unenhanced CT images of fat-poor angioliomyolipoma and RCCs. Specifically, they found that gray-level homogeneity and entropy were the strongest predictors of RCCs with an accuracy ranging from 83% to 93% [7]. Partly similar results were observed in present study that GLRLM gray-level nonuniformity normalized, shape flatness, and first-order uniformity were the top-ranked features for classifying ccRCCs versus fat-poor angioliomyolipomas and oncocytomas; in contrast, our results showed that radiomic score was excluded from the final radiologic-radiomic ML model against age, RE, and pseudocapsule. Younger age, absence of pseudocapsule, and lower RE are features associated with a diagnosis of benign fat-poor angioliomyolipomas and oncocytomas. The radiologic-radiomic ML model achieved an accuracy of 85.8%, which is comparable to the results reported by Hodgdon et al. and is better than the accuracies of the expert-level radiologist (73.5–95.1%).

The ability to differentiate benign fat-poor angioliomyolipomas and oncocytomas

Radiologic-Radiomic ML Analysis of Solid Renal Masses

TABLE 4: Diagnostic Performance of Radiologic-Radiomic Machine Learning (ML), Radiomic Score, and Expert-Level Radiologists in Evaluation of Three Disease Groups

Method	Sensitivity	Specificity	PPV	NPV	Accuracy	F1 Score
ccRCCs vs pRCCs and chrRCCs						
Radiomic score	154/190 (81.1)	53/64 (82.8)	154/165 (93.3)	53/89 (59.6)	207/254 (81.5)	304/355 (85.6)
Radiologic-radiomic ML	171/190 (90.0)	57/64 (89.1)	171/178 (96.1)	57/76 (75.0)	228/254 (89.8)	342/368 (92.9)
Reader 1	147/190 (77.4)	46/64 (71.9)	147/165 (89.1)	46/89 (51.7)	193/254 (76.0)	294/355 (82.8)
Reader 2	140/190 (73.7)	33/64 (51.6)	140/171 (81.9)	33/83 (39.8)	173/254 (68.1)	280/361 (77.6)
Reader 3	184/190 (96.8)	31/64 (48.4)	184/217 (84.8)	31/37 (83.8)	215/254 (84.6)	368/407 (90.4)
Reader 4	174/190 (91.6)	46/64 (71.9)	174/192 (90.6)	46/62 (74.2)	220/254 (86.6)	348/382 (91.1)
ccRCCs vs fat-poor angioleiomyolipomas and oncocytomas						
Radiomic score	116/190 (61.1)	24/36 (66.7)	116/128 (90.6)	24/98 (24.5)	140/226 (61.9)	232/318 (73.0)
Radiologic-radiomic ML	164/190 (86.3)	30/36 (83.3)	164/170 (96.5)	30/56 (53.6)	194/226 (85.8)	328/360 (91.1)
Reader 1	147/190 (77.4)	19/36 (52.8)	147/164 (89.6)	19/62 (30.6)	166/226 (73.5)	294/354 (83.1)
Reader 2	140/190 (73.7)	27/36 (75.0)	140/149 (94.0)	27/77 (35.1)	167/226 (73.9)	280/339 (82.6)
Reader 3	184/190 (96.8)	31/36 (86.1)	184/189 (97.4)	31/37 (83.8)	215/226 (95.1)	368/379 (97.1)
Reader 4	174/190 (91.6)	32/36 (88.9)	174/178 (97.8)	32/48 (66.7)	206/226 (91.2)	348/368 (94.6)
pRCCs and chrRCCs vs fat-poor angioleiomyolipomas and oncocytomas						
Radiomic score	48/64 (75.0)	29/36 (80.6)	48/55 (87.3)	29/45 (64.4)	77/100 (77.0)	96/119 (80.7)
Radiologic-radiomic ML	47/64 (73.4)	33/36 (91.7)	47/50 (94.0)	33/50 (66.0)	80/100 (80.0)	94/114 (82.5)
Reader 1	39/64 (60.9)	32/36 (88.9)	39/43 (90.7)	32/57 (56.1)	71/100 (71.0)	78/107 (72.9)
Reader 2	31/64 (48.4)	27/36 (75.0)	31/40 (77.5)	27/60 (45.0)	58/100 (58.0)	62/104 (59.6)
Reader 3	33/64 (51.6)	31/36 (86.1)	33/38 (86.8)	31/62 (50.0)	64/100 (64.0)	66/102 (64.7)
Reader 4	18/64 (28.1)	32/36 (88.9)	18/22 (81.8)	32/78 (41.0)	50/100 (50.0)	36/86 (41.9)

Note—Data are reported as number of lesions/total number of lesions (%). PPV = positive predictive value, NPV = negative predictive value, F1 score = balanced F score, ccRCC = clear cell renal cell carcinoma, pRCC = papillary renal cell carcinoma, chrRCC = chromophobe renal cell carcinoma.

mas from atypical RCCs such as pRCCs and chrRCCs is clinically significant because there are significant differences in the management and prognosis between benign and malignant lesions [29]. Fat-poor angioleiomyolipomas and oncocytomas, even though they are a small proportion of all renal neoplasms, often can be mistaken for RCCs on imaging. Distinguishing fat-poor angioleiomyolipomas and oncocytomas from pRCCs and chrRCCs is often difficult for expert-level radiologists; as shown in our results, the accuracy of expert-level radiologists ranged from only 50.0% to 71.0%, which is slightly better than midway between random chance and perfect discrimination. The radiologic-radiomic ML model using pseudocapsule, RE, and radiomic score can achieve an accuracy of 80.0% for lesion classification. These findings revealed strengths of radiologic-radiomic ML measurements when compared with routine clinical expert-level radiologist assessment. Some of our findings are consistent with the results of a study using a simi-

lar radiomic analysis in CT-detectable renal masses [20].

A couple of limitations of our study warrant mention. First, we restricted our study to the analysis of radiomic features on corticomedullary phase CT images. There are few reports available in the literature regarding the effects of using nephrographic phase CT images for radiomic analysis. However, we cannot rule out the possibility that corticomedullary and venous phase CT images are depicting different tumor characteristics; this possibility could represent an opportunity for future research. Second, another limitation of this study was the relatively small sample size, specifically the sample sizes for pRCC, fat-poor angioleiomyolipoma, and oncocytoma, reflecting overall low incidence of those renal mass subtypes. Finally, the radiomic and ML model analyses included in our study were selected before fold cross-validation because of limited sample size for each subgroup. In further study, the model can be internally and exter-

nally validated using the other database to investigate its reproducibility.

Conclusion

Our results show that routine expert-level radiologist interpretation of CT images had obviously large variances with relatively low accuracy in differentiation of benign and malignant solid renal masses, whereas the radiologic-radiomic ML approach provides an assessment of their ability to aid standardization of CECT interpretation. Our radiologic-radiomic ML model, comprising quantitative radiomic features and a priori radiologic hallmarks that are different from a DL black-box algorithm, was able to significantly reduce the misclassification of renal mass lesions. Considering the interpretability of our radiologic-radiomic ML model, we believe that the radiologic-radiomic ML approach could be a potential adjunct to expert-level radiologist interpretation of CT images for improving interreader concordance and diagnostic performance in routine clinical assessment of renal masses.

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